Enterprise Data Platform - Data Quality Documentation

Comprehensive Layer-by-Layer Data Specifications & Governance Manual

Generated Date: 2026-10-11 00:58:45	**Architecture:** Medallion (Bronze / Silver / Gold)
Engine: Apache Spark 4.1.2 & Keyword DQ Engine	**Governance Status:** **97.78% Overall DQ Score (EXCELLENT)**

1. Medallion Architecture Layer & File Format Matrix

The platform ingests multi-source higher education datasets and processes them through three rigorous Medallion stages. To support human inspectability alongside enterprise analytical performance, layer formats are configured as follows:

Layer	Storage Format	Target Directory	Purpose & Audit Metadata
Source	CSV	`data/source/*.csv`	Raw external institution extracts without alteration.
Bronze	**CSV** (Clean standalone CSVs)	`data/bronze/*.csv`	1:1 Exact Ingestion copy. Adds _bronze_ingestion_ts and _bronze_source_file. Enables direct inspection in CSV viewer/Excel.
Silver	**CSV & Parquet**	`data/silver/*.csv`	Cleansed, standardized, deduplicated datasets passing all DQ checks. Adds _silver_processed_ts, _pipeline_version, _is_valid=True.
Gold (SCD2)	**CSV & Parquet**	`data/gold/dim_student_scd2.csv`	Slowly Changing Dimension Type 2 (dim_student_scd2) with version, effective_start_date, effective_end_date, is_current.
Gold (Wide Mart)	**CSV & Parquet**	`data/gold/fact_student_enrollment_wide.csv`	18-column consolidated analytical mart (fact_student_enrollment_wide) combining all 4 entities for BI dashboards.

2. Layer 1 & 2: Source Ingestion & 4-Way Bronze Reconciliation

Every ingested dataset undergoes automated statistical data profiling and 4-way Source-to-Bronze cryptographic reconciliation to guarantee zero data loss, exact schema fidelity, and cryptographic bit-level payload consistency:

Dataset	Source Rows	Bronze Rows	Schema Match	Count Match	SHA-256 Checksum	Numeric Agg
Institution	6	6	PASS (100%)	PASS (Δ=0)	MATCH	N/A
Program	9	9	PASS (100%)	PASS (Δ=0)	MATCH	N/A
Student	17	17	PASS (100%)	PASS (Δ=0)	MATCH	N/A
Enrollment	18	18	PASS (100%)	PASS (Δ=0)	MATCH	PASS (Academic_Year)

3. Layer 3: Silver Cleansing, Validation & Quarantine Routing

Silver layer applies standardized domain cleansing (gender mapping, string trimming, ISO date formatting) and executes declarative rules across 6 DQ dimensions. Valid records proceed to Clean Silver; records violating rules are non-destructively isolated into Silver Quarantine with explicit dq_errors diagnostic arrays:

Entity	Total Ingested	Silver Clean	Quarantined	Primary Root Causes of Quarantined Records
Institution	6	**6** (100%)	0	All records fully valid and complete.
Program	9	**9** (100%)	0	All foreign keys reference active institutions.
Student	17	**10** (58.8%)	**7** (41.2%)	• STU005: Invalid Emirates ID regex format. • STU006, STU012: Non-existent Institution Code FK. • STU013: Unrecognized Gender domain. • Null PK row: Missing Student_ID. • STU002, STU014: Superseded duplicate versions. • STU_DQ_07: Cross-key duplicate Full_Name & Emirates_ID across different Student_IDs.
Enrollment	18	**11** (61.1%)	**7** (38.9%)	• ENR005: Non-existent Student FK (STU_NON_EXISTENT_999). • ENR006: Chronology anomaly (Graduation 2022 before Enrollment 2023). • ENR007: Orphan Program Code. • ENR013: Missing Student_ID. • ENR015: Orphan Institution. • ENR016: Historical Academic Year outlier (1990). • ENR017: Invalid Status domain.

4. Layer 4: Gold Consumption Layer Models

The Gold layer delivers two specialized analytical representations for downstream BI reporting and data science:

A. Slowly Changing Dimension Type 2 (dim_student_scd2)

Maintains full temporal lineage for student entity changes. Inactive versions are stamped with closing timestamps; current profiles are assigned is_current=True and effective_end_date='9999-12-31 23:59:59'.

Student_ID	Full_Name	Gender	Effective Start	Effective End	Current?	Ver
STU001	Fatima Ahmed Al-Mansoori	F	2023-01-15 08:30:00	9999-12-31 23:59:59	TRUE	1
STU002	Zayed S. Al-Nahyan	M	2023-04-11 10:00:00	9999-12-31 23:59:59	TRUE	1
STU003	John Robert Smith	M	2023-01-17 11:00:00	9999-12-31 23:59:59	TRUE	1
STU004	Aisha Yusuf Mohammed	F	2023-01-18 10:45:00	9999-12-31 23:59:59	TRUE	1
STU014	Carlos Silva Junior	M	2023-05-20 15:00:00	9999-12-31 23:59:59	TRUE	1

B. 4-Table Consolidated Analytical Mart (fact_student_enrollment_wide)

Denormalizes Student, Enrollment, Institution, and Program dimensions into a unified 18-attribute analytical table. Enforces Gold-layer business rules (Active Institution, Active Program, Minimum Age >= 16).

Enrollment_ID	Student_Name	Institution_Name	Program_Name	Status	Mart_Version
ENR001	Fatima Ahmed Al-Mansoori	Emirates National University	BSc Computer Science	Enrolled	Gold-Wide-v1

ENR002	Zayed S. Al-Nahyan	Emirates National University	Master of Business Admin	Enrolled	Gold-Wide-v1
ENR003	John Robert Smith	Gulf Science Academy	BEng Mechanical Engineering	Enrolled	Gold-Wide-v1
ENR004	Aisha Yusuf Mohammed	Gulf Science Academy	Doctor of Medicine	Graduated	Gold-Wide-v1
ENR008	Sarah Jane Connor	Dubai Future Polytechnic	MSc Artificial Intelligence	Enrolled	Gold-Wide-v1

5. Data Quality Dimensions Health Scorecard

Overall platform health score is calculated across 36 automated checks spanning all 6 standard DQ dimensions:

DQ Dimension	Score (%)	Grade / Status	Evaluated Checks & Key Coverage
Completeness	**99.12%**	EXCELLENT	Source schema reconciliation, record counts, mandatory primary keys (Student_ID, Enrollment_ID, Institution_Code, Program_Code).
Consistency	**99.21%**	EXCELLENT	Row-level SHA-256 payload checksums, cross-table status compatibility, Academic_Year logical bounds.
Validity	**97.14%**	EXCELLENT	Emirates ID standard regex (784-YYYY-XXXXXXX-Z), Institution Code format, allowed status domains.
Referential Integrity	**94.31%**	GOOD	Enrollment -> Student/Program/Institution foreign key verification; Program -> Institution parent checks.
Uniqueness	**94.12%**	GOOD	Window partition natural key deduplication on Student_ID and cross-key Emirates_ID + Full_Name duplicate isolation.
Timeliness	**100.00%**	EXCELLENT	Chronology validation: Updated_Date >= Created_Date, Graduation_Date >= Enrollment_Date.
OVERALL SCORE	**97.78%**	**EXCELLENT**	**36 Automated Rules Evaluated across Bronze, Silver, and Gold Layers**

6. Data Quality Action & Routing Hierarchy

The platform enforces a multi-tier governance hierarchy separating fatal system errors from business record anomalies:

Action	Severity	Execution Behavior	Layer & Destination
REJECT / FAIL	**CRITICAL**	Aborts batch execution immediately with DataQualityException.	Bronze Ingestion (Schema mismatch, Record loss, Checksum failure).
QUARANTINE	**CRITICAL / HIGH / MED**	Non-blocking row-level isolation. Enriches records with dq_errors diagnostic metadata array.	Silver Quarantine (data/silver/quarantine/) & Gold Mart Quarantine.
WARN / PASS	**LOW**	Audit log warning emitted; allows record to pass through to clean datasets.	Informational metrics, minor formatting conventions.